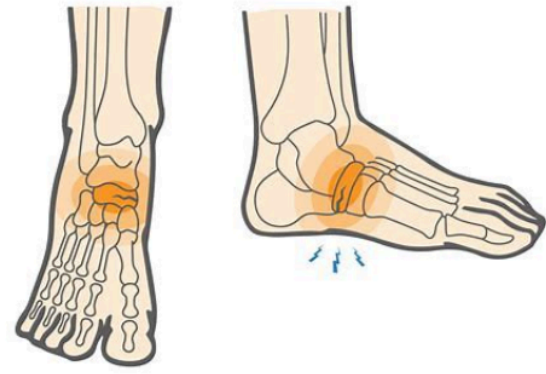


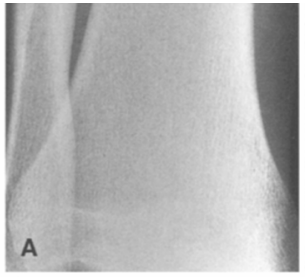
Stress Fractures

Presented by Elisheva Shiri Decktor

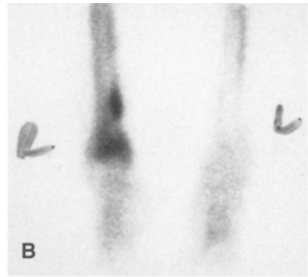


What's Inside ^[5]

A Conventional radiograph shows subtle areas of decreased cortical density.



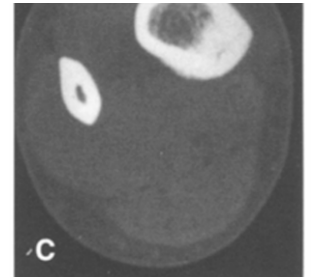
3 weeks after shows focal, unilateral, vertically oriented area of increased uptake.



4 months after shows endosteal and periosteal new bone formation with a linear line.



9 months after shows focal remodeling of distal tibial medial cortex.



Challenge Proposal:

Stress fractures, prevalent in athletes[1] and combat military soldiers[2], cause damage to bones and surrounding tissues[3]. The continuous use of an unhealthy bone hampers its natural healing process through disruption of ongoing bone remodeling [4][5]. Repeated activation on top of existing fractures impedes adequate mineralization and matrix formation by osteoblasts and osteoclasts[6]. Non-invasive physiological examinations[7], X-ray tests [8], CT or MRI[9] facilitate the most common diagnosis today. However, they rely on self-diagnosis, mainly related to pain, after significant damage occurs [10].

The need for real-time, non-invasive analysis is crucial. LoC models demand miniaturized, rapid, and automated testing processes that need to operate with small sample volumes and provide real-time analysis of a target analyte[11].

A possible analyte to target is the alkaline phosphatase (ALP), an essential biomarker associated with bone-related diseases such as fractures. Current technologies for ALP detection include colorimetry, chemiluminescence, fluorescence, and electrochemical sensors. However, they suffer low enhanced sensitivity on small scales such as biosensors and low specificity and efficiency, particularly in actual serum samples, to accurately assess ALP levels for effective diagnosis and treatment monitoring of bone-related conditions[12].

Another option is the analyte-targeted osteocalcin, a protein associated with bone metabolism, which is a biomarker for the presence and progression of stress fractures. Osteocalcin can be tested using various methods such as enzyme-linked immunosorbent assay (ELISA), radioimmunoassay (RIA), and chemiluminescent immunoassays. However, these conventional methods require relatively large sample volumes, involve complex procedures, and often entail time-consuming steps, making them less suitable for integration into Lab-on-a-Chip (LoC) models[13].

ALP levels remain stable in men throughout life; however, the ALP level in women increases around menopause [14]. The osteocalcin level is low up to middle age in men and increases afterward; however, in women, it follows the same pattern as in ALP [15]. The optimal choice for detecting bone stress fractures on a Lab-on-a-Chip (LoC) is APL rather than osteocalcin.

Novel biosensing technologies tailored for miniaturization, sensitivity, and user-friendliness are necessary to implement ALP detection on a LoC. These biosensors should detect osteocalcin in small sample volumes, offer real-time analysis, and operate with high sensitivity and specificity. Developing such technologies is essential for accurately detecting osteocalcin in a miniaturized, sensitive, and user-friendly manner on LoC platforms. With that, correct early detection of stress fractures will prevent fatal injuries and avert complete activity shutdown[16][17].

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